Naturalness is history

This paper addresses the idea of *naturalness* from the perspective of language as complex system. In the historical domain, sound changes are considered natural if they are phonetically motivated, this idea is often extended to synchrony where phonological rules are also modelled on the basis of naturalness (Donnegan & Stampe, 1979; Kiparsky, 1982; Blevins, 2004a, 2006). I model language as complex dynamic system (CDS) and argue that it is the *gliders* (self-similar patterns) within this system that constitute natural sound changes. As such, I argue that natural sound changes reflect properties of emergent and external system which cannot be reduced to a representation within the I-language phonology of a speaker.

I use the term *H-language* refers to language as a CDS. The H-language is the emergent product of the iterated transmission of I-languages and exists in a supervenience relationship with elements at the substrate level (I- and E-language), meaning that although H-language derives its properties from these elements it is not reducible to or caused by any single substrate element (List & Spiekermann, 2013). As H-language emerges from transmission its properties are determined by the constraints acting on the linguistic module of the mind (i.e., the shape of a possible I-language) and by the constraints that govern the externalisation of I-language (e.g., domain-general cognitive processes, physiological constraints, etc).

Gliders arise in a CDS when some properties of the system, *control parameters,* exert some pressure on the system shifting it towards a certain state. The combined action of multiple control parameters can produce gliders that iterate over time or "move" through the system (Rickles et al, 2007). I argue that "natural" sound changes are examples of such gliders in H-language and result from the interaction of three control parameters:

1. Phonetic motivation
2. Directionality
3. High frequency input

1) Phonetic grounding is necessary as only those changes motivated by universal properties stand the chance of being common enough to emerge as gliders. 2) In the process of emergence, a positive-feedback-loop is created which reinforces strongly directional changes, like /p/ > /f/. Whereas changes where either direction is approximately equally likely cancel each other out and so are much less likely to appear on the emergent level. 3) Patterns of change that affect common sounds have more chance of occurring and therefore are more likely to show up as gliders in the emergent system.

Secondly, the nature of the supervenience relationship between substrate and emergent level means that it is not possible to determine a direct causal path from a substrate-level event to a property of the emergent product. Variations in the relations between phonetic motivations and other parameters in the system allow this permanent factor to account for specific changes because although physiological pressures are permanent their effect size is not constant – thus, Bloomfield's (1993: 386) is concern mitigated.

Thus, the frequency of certain, so-called natural sound changes can be accounted as features of the emergent system. Natural sound changes are indeed phonetically motivated and typologically common but the latter is determined by the factors governing emergence in complex systems. Having shown how natural processes exist in a supervenience relationship with I-language and the factors governing its transmission it becomes clear that natural processes can be explained in the external domain without the need to be encoded in the I-language.